

11 LAND: THE ULTIMATE SCARCE RESOURCE

Did you ever pause to think how strange it is that a person from the Netherlands wrote this book about how solar panels will become our main source of energy?

Did you know that 95% of the people in the world get more sun than we Dutchies do?

The further away from the equator, the less sun year round, especially in winter. The Netherlands is very far from the equator. On top of this we are also one of the most densely populated countries in the world with a lot of energy intensive industry: greenhouses, data centers, chemical plants. In short we are uniquely *unsuited* for solar energy. For this reason, the most cost effective solution for the Netherlands is one in which wind energy takes the lead. I expect that we will get about 70% of our energy from wind turbines, dotted over about 10% of our part of the North Sea. The turbines themselves, including their foundations, occupy about 0.01 percent of that sea area. The rest is open water. Fishing boats can still fish. Ships can still sail. The small artificial reefs around the turbine foundations actually increase biodiversity compared to the sandy bottom that was there before. Ecologists have documented more fish, more crabs, more seabirds in the wind farm zones. I utterly fail to see the land-use problem of wind.

For most of the world, the situation is very different. They live in places with more sun than we have. I estimate that for 90 percent of the global population, solar panels on rooftops with modest battery storage is the most cost-effective solution. How much space would that take?

11.1 The Surprising Arithmetic of Solar Land Use

Consider a future in which every person on Earth uses as much energy as the average American. Not the average European, who gets by with half. Not the average Indian, who uses far less still. The average American, who uses a luxurious four times more than the average world citizen.

Americans are not exactly famous for frugality. They drive large cars, air-condition large houses, and run more appliances than anyone. On top of that you have industry, flying, building... We are going to replace all of that with solar panels. The whole shebang needed to power the country. Can everybody on earth have that much energy?

If we produced all that using solar panels, it would be about 75 kilowatt-hours per person per day.

Three-quarters of the world's population lives in places that receive about 4 kilowatt-hours of solar energy per square meter per day, averaged over the year. Current solar panels today convert 20% to 25% of that into electricity. Their efficiency is still increasing constantly, but let's conservatively take 25% efficiency. Then each square meter of solar panel produces about 1 kilowatt-hour per day.

That is a nice number to remember: most solar panels will produce about 1 kilowatt-hour per square meter per day.

That means that for 75 kilowatt-hours per person per day, you need 75 square meters of solar panels.

That might sound like a lot until you start comparing.

In developed countries, the roof space per person alone is 75 square meters. The built environment, including roads, parking lots, driveways, commercial buildings, and industrial facilities, is about 750 square meters per person. So we're talking about solar panels covering 10 percent of the built environment. Or, if you prefer a more intuitive measure: roughly one and a half parking spaces per person.

At the global scale, 75 square meters per person across 10 billion people would be about 0.4 percent of Earth's land surface. Less than half a percent. The world already has more than that in paved roads and parking lots. I think we'll manage.

The objection that solar takes up too much space becomes even more absurd when you compare solar to what we already have. Open pit coal mines leave scars visible from space. Mountaintop removal in Appalachia has leveled hundreds of peaks and buried thousands of miles of streams. The tar sands of Alberta have stripped boreal forest from an area larger than some European countries. Oil extraction requires drilling pads, access roads, pipelines, refineries, storage tanks, and tanker ports. Natural gas needs much of the same. The fossil fuel system already has a sizable land footprint too. We just stopped noticing it.

SIDEBAR: We Have Room

Land use of solar vs. built environment

What	Area
Solar panels needed per person (US-level consumption)	~75 m ²
Existing roof space per person (developed countries)	~75 m ²
Total built environment per person	~750 m ²
Solar as fraction of built environment	10%
Solar as fraction of all land	0.4%

Solar panels fit on existing roofs. We have ten times more built environment than we need for solar. We don't have to take anything away from nature or agriculture.

Land use of solar vs agriculture

What	Fraction of habitable land
Agriculture	60% (needs fertile soil)
Nature	37% (needs fertile soil)

What	Fraction of habitable land
Built environment	3% (1% 'impermeable infrastructure')
Solar needed for all energy	0.4% (doesn't need fertile soil)

11.2 The Myth That Won't Die

Once you know these numbers, you might wonder why so many people keep insisting that solar and wind cannot work. The claim is usually that their "energy density" is too low.

The argument goes something like this: fossil fuels and nuclear pack enormous energy into small spaces, while solar and wind are diffuse, spread thin across the landscape. Furthermore, the argument continues, history has always moved toward denser energy sources, and solar and wind represent a step backward.

Vaclav Smil, whom I consider a great energy historian, has made versions of this argument for decades. He believes there is a kind of natural law pushing civilization toward higher energy density. This implies nuclear power must follow fossil fuels because nuclear produces more energy per square meter than anything else.

There's just one problem. History has already falsified his prediction. We have been choosing solar and wind over nuclear for four decades now. Not because politicians or environmentalists forced it, but because markets did. Solar and wind rode their learning curves downward while nuclear rode it upward. Whatever "law" Smil thought he discovered, the world decided to ignore it. He is a great energy historian, but a great energy futurist he is not.

The energy density argument persists anyway, pushed by three groups who have accidentally formed an unholy trinity.

First, we have the fossil fuel lobbyists. This will surprise exactly nobody. Armed with deep pockets and a playbook borrowed from the tobacco industry they spread fear, uncertainty, and doubt (FUD). They will not hesitate to make solar seem impractical on any dimension, whether it's true or not.

Second, nuclear proponents. I interact with them constantly, and many are smart, sincere, and genuinely worried about climate change. They believe that land use is the killer argument against solar, the one that will finally convince people that nuclear is the only serious option. And they have a point, narrowly speaking: nuclear does produce about 100 times more electricity per square meter than solar.

But 100 times more than a tiny number is still a small number. If solar requires 75 square meters per person and we have 750 square meters of built environment per person, then solar "uses" 10 percent of an area that can't be used for nature or food anyway. And you can't build a nuclear plant on your roof.

Third, farmers and conservationists. This is the group that puzzles me most, because they have it exactly backwards. They worry that solar farms will pave over agricultural land and

nature reserves. They think they're defending the Earth, but they are attacking it. And they are worried about the wrong thing entirely.

11.3 The Real Land Hog

About 60% of all habitable land on Earth is used for agriculture. Nature comes in as a distant second with 37%. The total build environment is about 3% of which 1% is actually paved over.

Within that agricultural land, about three quarters is devoted to livestock: pastures for grazing, and cropland for growing animal feed. The other quarter grows food that humans eat directly. That quarter produces 80% of calories and 60% of protein.

A single beef burger, containing perhaps 25 grams of protein, requires about 40 square meters of land to produce. Two burgers per year require more land than solar panels to cover all your energy needs. Not just your electricity needs. Not just your home energy needs. *All* your civilizational energy needs. Two burgers.

If you're worried about humanity's land footprint, the solar panel on your roof is the last thing you should worry about.

11.3.1 Sidebar: The Real Land Hog

Meat's Land Footprint

Food	Land use per 100g protein
Beef	164 m ²
Lamb	185 m ²
Pork	11 m ²
Poultry	7 m ²
Eggs	6 m ²
Tofu	2.2 m ²
Pulses	1.5 m ²
Microfermentation	0.3 m ²
Future Food	0.03 m ²

Beef requires over a 100 times more land than pulses for the same amount of protein. Two burgers per year requires about 80 m² of land. That's a bigger area than the solar panels that care of *all* your energy use. And you can't put a cow on the roof.

Microfermentation can take sugar from sugarcane and let micro-organisms convert it to protein. You need about 250g of sugar for 100g of protein and you can produce that in a year with 3m². You also need about 0.1 kWh per 100g protein to power the bioreactor but the amount of solar panel for that relatively insignificant.

For Future Food (which is still in the laboratory) I've taken the example of Solafood who produce protein by feeding micro-organisms hydrogen and CO₂ that they produce with energy from solar panels. You need 50g of hydrogen produced with 2.5 kWh of electricity. You also need 500g of CO₂ captured with (in the future) 5 kWh of electricity. Bioreactor operation, water treatment, etc. takes another 2 kWh. About 10 kWh per 100g protein in total. That means about 0.03 m² of solar panels on your roof. Cool right?

Solar vs agriculture: energy per square meter

Solar panels	1 kWh per m ² per day
Average cropland (food calories)	0.005 kWh per m ² per day

Solar produces roughly 200 times more energy per square meter than agriculture.

Why? Plants are beautiful but inefficient. Crops convert roughly 0.5% of solar energy into biomass. And most of that is inedible: roots, stems, leaves. What ends up on your plate (or in your tank as ethanol or biodiesel) is less than 0.1%. Solar panels convert 20-25% percent of sunlight into electricity and their efficiency is still increasing.

It becomes even worse if you feed that plant to an animal. By the time a calorie of sunlight becomes a calorie of beef, 99.99% has been lost along the way.

11.4 What We Could Give Back

Solar panels use sunlight over 200-times more efficient than crops. Over 10 thousand-times more efficient than beef.

These are not just interesting facts. This is what a rational energy land use discussion should be about.

Precision fermentation can produce proteins from microbes growing in tanks. Cultivated meat grows real animal muscle without the animal. Plant-based alternatives are improving so fast that for an increasing amount of products (like burgers) most taste testers can't tell the difference.

These technologies are still young and expensive but they are riding down the learning curve. Which means costs will fall, quality will rise, adoption will accelerate, which will make costs fall further... a virtuous cycle.

There is no physical reason that prevents us from producing protein that is healthier and tastier than beef. No reason that prevents us from doing that with a thousand times less land. (See sidebar.)

Then we can really start giving back to nature. Currently 60% of habitable land is being claimed by agriculture. If we replace animal protein production by something smarter we not only reduce animal suffering but we could also give back two thirds of agricultural land back to nature. If you care about nature, that's a lot more interesting than 0.4% of land that is already paved over.

11.4.1 Sidebar: Rewilding Success Stories

Oostvaardersplassen, Netherlands

What was once farmland became a 56-square-kilometer nature reserve in the 1970s. Wild horses, red deer, and Heck cattle now roam grasslands and wetlands. Rare bird species returned. Ecologists debate management methods, but the land transformation is undeniable.

Iberá Wetlands, Argentina

Once drained for cattle, now South America's largest rewilding project. Jaguars, giant anteaters, and pampas deer have been reintroduced. Ecotourism now generates more income than the ranching it replaced.

Scotland's Cairngorms

Scots pine forest is returning to the Highlands after centuries of sheep grazing. Native species are recovering. Red squirrels, capercaillie, and pine martens are spreading.

Yellowstone, USA

Wolf reintroduction in 1995 triggered ecosystem-wide changes. Elk behavior shifted, allowing riverbank vegetation to recover. Stream channels stabilized. Songbird populations increased. One species returned, and the land transformed.

European Bison

Nearly extinct in the wild by 1927, European bison now number over 7,000 across Poland, Belarus, Romania, and elsewhere. Rewilding projects are creating new herds and restoring grazing ecosystems.

When we stop using land, nature often comes back faster than we expect.

11.5 The Final Frontier

"Space, the final frontier." If you are a Star Trek fan like me you can hear these words in your head. The crew of the Starship Enterprise would go and discover new worlds everywhere in the universe. But for the foreseeable future, all life in the known universe is here on earth.

If I see a picture of earth from space, if I look at that blue marble, I feel wonder, awe, gratitude, responsibility.

Most life on earth lives in a band going from just a few meters below the ground to a couple of dozen meters above the ground. Scaled to a globe with a diameter of 30 cm, that's just a few micrometres. This incredibly thin layer, this surface tension, fed by the energy from the sun, is the ultimate scarce resource. The final frontier. And yet we are turning it into monocultural agricultural deserts. Overfeed it with fertilizer. Spray it with poison to kill all life we don't want. I understand we need to eat. Over 2 billion people in the world often don't have enough to eat. But let's try to feed ourselves efficiently and in harmony with nature.

If you are worried about land use, look at your plate, not your roof. So that concludes our deep dive in all the problems that could keep us from abundant energy. Time to wrap up using the framework called planetary boundaries.